

CAF Current State — Vol 4: Quality, Ops & Agent Map

Aggregated export — 2026-07-10. Source markdown in repo is unchanged.

Bundle id: 14-caf-current-state-vol4-ops

Included:

docs/volumes/CAF_CONTEXT_VOL4_Quality_Ops_and_Agent_Map.

CAF Current State Context Pack — Quality Ops and Agent Map

Volume 4 of 4 | Generated 2026-07-10 | Full pack: docs/CAF_CURRENT_STATE_CONTEXT_PACK.md

12. QC, risk, validation, and quality gates

QC runtime (qc-runtime.ts → runQcForJob)

1. **Checklist** from flow_definitions.qc_checklist_name → qc_checklists
2. **Copy quality** patterns (hashtags, tone, etc.)
3. **risk_policies** — global + flow-scoped via listRiskPoliciesForJob()
4. **brand_constraints.banned_words** — same keyword scan as policies

NOT enforced by QC

- **risk_rules** (project CSV rows) — zero references in qc-runtime.ts
- Honesty endpoint: GET /v1/projects/:slug/risk-qc-status

Pass/fail

qc_passed = no blocking checklist failures AND no CRITICAL risk findings.

Routing (validation-router.ts)

Maps recommended_route to job status; finalJobStatusAfterRender() → IN_REVIEW .

Other gates

- **Output schema validation** — CAF_OUTPUT_SCHEMA_VALIDATION_MODE (off | warn | strict)
- **Draft package contract** — CAF_DRAFT_PACKAGE_CONTRACT_MODE
- **Diagnostic audits** — machine quality evaluation
- **Auto-validation** — auto_validation_results table
- **Post-approval LLM review** — llm_approval_reviews , upstream recommendations via insertLlmApprovalReview()
- **Human review gate** — CAF_REQUIRE_HUMAN_REVIEW_AFTER_QC=true default

13. Learning loop

Facade (mandatory entry point)

src/services/learning-rule-selection.ts

Function	When	Rules
getLearningRulesForPlanning()	Decision engine scoring	BOOST_RANK, SCORE_BOOST, SCORE_PENALTY only
getLearningContextForGeneration()	LLM prompt build	generation family / GUIDANCE action types

Planning path

decision_engine/index.ts → applyLearningBoosts() multiplies pre_gen_score .

Sources: editorial-learning.ts , performance-learning.ts , market-learning.ts .

Generation path

learning-context-compiler.ts → merged_guidance injected in llm-generator.ts .

Preview: GET /v1/learning/:slug/context-preview .

Global rules disabled in compiler.

Evidence → rules loops

Loop	Trigger	Output
Editorial (B)	POST .../editorial-analysis	Pending SCORE_PENALTY rules
Performance (C)	performance ingest/CSV/analysis	Ranking rules from metrics
LLM approval	POST .../llm-review-approved	Upstream recommendations + hints
Manual	apply/retire/dismiss APIs	pending → active

Attribution / snapshots

- learning_generation_attribution — which rules affected a generation
- run_context_snapshots via setRunContextSnapshot() — prompt versions + fingerprints (failures logged, never abort run)
- job_outcomes (075) — publish → performance anchor

Tables

learning_rules , learning_observations , learning_hypotheses ,
learning_insights , performance_metrics , llm_approval_reviews

14. Publishing and integrations

Model

publication_placements — one row per scheduled post: caption/media snapshots, platform, format
(carousel | video | unknown).

Lifecycle

draft → scheduled → publishing → published | failed | cancelled

Executor modes (CAF_PUBLISH_EXECUTOR)

Mode	Behavior
none (default)	Start claims placement; returns n8n payload; external worker calls /complete

dry_run	Fake IDs for plumbing tests
meta	Core calls publishPlacementToMeta() directly

Meta integration (meta-graph-publish.ts)

Facebook feed/photos/video; Instagram single/carousel/reels; polls container status; re-signs Supabase URLs for Meta fetch.

Credentials: `project_integrations` (`META_FB` , `META_IG`) + env overrides `CAF_META_*` .
Account aliasing: `CAF_META_ACCOUNT_SOURCE_MAP` (e.g. `CUISINA=SNS`).

n8n payload

`publication-n8n-payload.ts` — legacy field names matching `Publish_Carousel_IG_FB` / `Publish_Video_IG_FB` .

On success

`appendPublicationResultToJob()` + `upsertJobOutcomeOnPublish()` → closes loop to `job_outcomes` / performance.

Project integrations

`src/routes/project-integrations.ts` — CRUD + test for external credentials.

15. API surface map

Registration hub: `src/server.ts`

Domain	Route file	Key endpoints	Auth
Health	<code>v1.ts</code>	<code>/</code> , <code>/health</code> , <code>/readyz</code> , <code>/health/rendering</code>	Optional bearer
Decisions / jobs	<code>v1.ts</code>	<code>/v1/decisions/plan</code> , <code>/v1/jobs/ingest</code> , GET <code>.../jobs/:slug/:task_id</code>	Token if required
Review queue	<code>v1.ts</code>	<code>/v1/review-queue/...</code> (large surface: <code>decide</code> , <code>regen</code> , <code>reprint</code> , <code>docai</code> , <code>mimic-mode</code>)	Token
Runs	<code>runs.ts</code>	CRUD, <code>/jobs</code> , <code>/start</code> , <code>/process</code> , <code>/render</code> , <code>/cancel</code>	Token
Pipeline	<code>pipeline.ts</code>	<code>/generate</code> , <code>/qc</code> , <code>/diagnose</code> , <code>/full</code> , <code>/rework</code>	Token
Projects	<code>project-config.ts</code>	<code>strategy</code> , <code>brand</code> , <code>risk-qc-status</code> , <code>flow-types</code> , <code>brand-assets</code> , <code>heygen</code>	Token
Flow engine	<code>flow-engine.ts</code>	<code>flows</code> , <code>prompts</code> , <code>schemas</code> , <code>templates</code> , <code>qc-checks</code> , <code>risk-policies</code>	Token
Signal packs	<code>signal-packs.ts</code>	<code>upload</code> , <code>ideas</code> , <code>select-ideas</code> , <code>mimic-mode overrides</code>	Token
Inputs evidence	<code>inputs-evidence.ts</code>	<code>upload</code> , <code>list</code> , <code>rows</code>	Token
Inputs scrapers	<code>inputs-scrapers.ts</code> , <code>inputs-sources.ts</code>	<code>workbook</code> , <code>scraper runs</code>	Token
Inputs processing	<code>inputs-processing.ts</code>	<code>profile</code> , <code>insight passes</code> , <code>build-signal-pack</code> , <code>idea-lists</code>	Token

Evidence read	evidence-insights-read.ts	/v1/evidence, /v1/insights	Token
Creative intel	creative-intelligence.ts	ingest, analyses, styling, mimic-template	Token
Learning	learning.ts	rules, observations, performance, editorial-analysis, llm-review	Token
Publications	publications.ts	CRUD, start, complete, n8n-payload	Token
Integrations	project-integrations.ts	Meta etc.	Token
Admin	admin.ts, mimic-text-overlay-lab-routes.ts	HTML workbench + lab APIs	Token
Renderer templates	renderer-templates.ts	/api/templates	Often open
Review proxy	review-proxy.ts	Non-Core paths → embedded Next.js	Review exempt

Auth: When `CAF_CORE_REQUIRE_AUTH` + `CAF_CORE_API_TOKEN`, Bearer or `x-caf-core-token` on Core paths.

16. Environment and deployment

Local minimum

```
DATABASE_URL=postgres://...
OPENAI_API_KEY=sk-...
# Optional for full loop:
RENDERER_BASE_URL=http://localhost:3333
CAF_CORE_URL=http://localhost:3847 # Review app
```

Production groups

See `ENV_AND_SECRETS_INVENTORY.md` and `src/config.ts` (Zod-validated).

Group	Notable vars
Core	DATABASE_URL, CAF_RUN_MIGRATIONS_ON_START, CAF_CORE_API_TOKEN, CAF_REVIEW_ENABLED, CAF_PUBLIC_URL
Planning	DECISION_ENGINE_VERSION, DEFAULT_MAX_CAROUSEL_JOBS_PER_RUN, score weights
QC	CAF_REQUIRE_HUMAN_REVIEW_AFTER_QC, CAF_OUTPUT_SCHEMA_VALIDATION_MODE
Rendering	RENDERER_BASE_URL, VIDEO_ASSEMBLY_BASE_URL, concurrency/ timeouts
Mimic	MIMIC_IMAGE_ENABLED, MIMIC_IMAGE_PROVIDER, BFL_API_KEY, MIMIC_LAYOUT_QA_*
Video	HEYGEN_API_KEY, SCENE_ASSEMBLY_CLIP_PROVIDER, SORA_*
Publish	CAF_PUBLISH_EXECUTOR, CAF_META_*, META_GRAPH_API_VERSION
Storage	SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY, SUPABASE_ASSETS_BUCKET
Inputs	APIFY_API_TOKEN, DOCUMENT_AI_*, CREATIVE_INTEL_*

Deploy

- **Core + Review:** `fly deploy -a caf-core` (Dockerfile builds Next.js standalone)

- **Renderer / media:** separate Fly apps (`fly.caf-renderer.toml` , `fly.toml`)
- **Migrations:** `npm run migrate` or on startup when enabled

17. Current maturity matrix

Feature / subsystem	Maturity	Evidence	Known gaps	Main files
Standard carousel	Production	Full pipeline + Review editor	Template variety	<code>carousel-render-pack.ts</code> , <code>CarouselEdits.tsx</code>
Carousel editor	Production	Live preview, brand styling panel	—	<code>CarouselSlider.tsx</code> , <code>preview-live-slide/route.ts</code>
Mimic carousel	Production (gated)	Full prep + render + layer editor	Default <code>MIMIC_IMAGE_ENABLED=false</code>	<code>mimic-draft-prep.ts</code> , <code>MimicCarouselLayerEditorPanel.tsx</code>
New visual carousel	Active development	<code>new-visual-carousel-*.ts</code> , migration 070/071	Docs lag	<code>new-visual-carousel-prep.ts</code>
Why Mimic carousel	Newer production	migration 074, SIL	Requires substantive SIL	<code>why-mimic-carousel-flow-types.ts</code>
BVS / brand bible	New (migration 078)	<code>brand-bibles.ts</code> , <code>bvs-v1.ts</code>	Docs not updated	<code>brand-bible.ts</code> , <code>bvs-render-plan.ts</code>
Mimic image (single)	Gated	<code>mimic-image-job.ts</code>	Same env gate	<code>mimic-image-provider.ts</code>
Mimic text placement automation	Partial	Layout QA loop only	Composite saliency not built	<code>mimic-post-render-layout-loop.ts</code>
HeyGen video	Production	<code>heygen-renderer.ts</code>	Provider quotas	<code>HEYGEN_API_KEY</code>
Scene assembly	Provider-dependent	<code>scene-pipeline.ts</code>	Sora availability	<code>sora-scene-clips.ts</code>
Inputs pipeline	Largely built	migrations 027–065+	Stage-3 picker partial	<code>inputs-processing.ts</code>
Creative intelligence	Production	migration 055	Downstream SIL docs thin	<code>creative-intelligence.ts</code>
Publishing	Production	Meta + n8n paths	Default executor none	<code>publications.ts</code>
Learning	Production	Facade + compiler	Global rules disabled	<code>learning-rule-selection.ts</code>
QC / risk	Production	<code>qc-runtime.ts</code>	<code>risk_rules</code> not enforced	<code>risk-qc-status.ts</code>
Review app	Production	Embedded in Core	Marketer funnel incomplete vs admin	<code>apps/review/</code>

Admin config	Production	admin.ts, project-config routes	HTML legacy mixed with API	project-config.ts
Product image flows	Placeholder	Blocked at LLM	Not wired	product-flow-types.ts

18. Repo map for future AI agents

Goal	Start here	Do not casually change
Run planning / scoring	src/decision_engine/, run-orchestrator.ts	ID patterns, planned_jobs_json contract
Job execution	job-pipeline.ts	Lifecycle enums
LLM generation	llm-generator.ts, flow-engine repos	Prompt versions without migration
QC	qc-runtime.ts	Write qc_result except via mergeGenerationPayloadQc
Risk policies	flow-engine risk-policies, qc-runtime.ts	Assume risk_rules are enforced
Rendering	job-pipeline.ts, carousel-render-pack.ts, services/renderer/	HeyGen retry without hasActiveProviderSession
Review UI	apps/review/src/components/TaskReviewClient.tsx	Job state authority
Mimic	mimic-draft-prep.ts, mimic-carousel-render.ts, mimic-payload.ts	Conflate carousel_package with mimic_carousel_package
BVS	brand-bible.ts, bvs-v1.ts, bvs-render-plan.ts	—
New visual	new-visual-carousel-prep.ts, new-visual-carousel-execution.ts	Re-add TP references to visual-first
Inputs	inputs-processing.ts, inputs-evidence.ts	Signal pack JSON contracts
Publishing	publications.ts, meta-graph-publish.ts	Placement lifecycle strings
Learning	learning-rule-selection.ts facade only	Direct listActiveAppliedLearningRules imports
Env / config	src/config.ts	—
Review contract	review-contract.test.ts	Rename paths without updating test
Schema truth	migrations/	—

19. Non-negotiable invariants

- task_id + project_id** remain primary execution keys across `job_drafts`, `assets`, `editorial_reviews`, `job_state_transitions`, etc.
- Postgres caf_core** is source of truth; Review app is a client.
- content_jobs.generation_payload** is a versioned integration contract — coordinate pipeline, Review, admin consumers.
- QC qc_result** must be written only via `mergeGenerationPayloadQc()` (`generation-payload-qc.ts`).

5. **HeyGen/provider retries** must use `hasActiveProviderSession()` — no double-submit when `video_id / session_id` exist.
6. **Learning lookups** go through `learning-rule-selection.ts` facade — not direct compiler imports from new code.
7. **risk_policies + brand banned_words** are QC-enforced; **project risk_rules** are **NOT** (see `risk-qc-status`).
8. **Lifecycle status strings** must not be casually renamed (runs have SQL CHECK; jobs do not but consumers depend on strings).
9. **Migrations + source win** over docs when conflicting.
10. **mimic_carousel_package** is for TP-grounded carousel render flows only (`isTpGroundedCarouselRenderFlow()`); standard carousel uses `carousel_package` .
11. **Mimic LLM creation pack** must filter signal pack to the job's single planned idea (`mimicFlowOnly: true`).
12. **Mimic carousel text** is overlay-only — do not bake copy into Flux at render (enforced in `job-pipeline.ts`).
13. **Upstream recommendations** must use `parseUpstreamRecommendations` + `insertLlmApprovalReview()` ; each item logged as `learning_observation` .
14. **Run context snapshots** via `setRunContextSnapshot()` — failures logged, never abort run.
15. **Review /v1/ contract** — changes to listed paths require `review-contract.test.ts` update in same PR.

20. Documentation drift and required updates

Still accurate

- `AGENTS.md` — invariants match code
- `docs/DOMAIN_MODEL.md` — ID patterns (minor column naming caveat)
- `docs/LIFECYCLE.md` — run states (job diagram omits `GENERATED`)
- `docs/QUALITY_CHECKS.md` , `docs/RISK_RULES.md` — `risk_rules` vs `risk_policies` asymmetry
- `docs/EXTERNAL_CONTEXT_PACK.md` — tier structure (needs new pack reference)
- `docs/CAF_INPUTS_PIPELINE_ROADMAP.md` — largely matches implementation
- `.cursor/rules/caf-domain-model.mdc` , `mimic-carousel-package.mdc` , `mimic-signal-pack-llm-filter.mdc`

Outdated or incomplete

Doc	Drift
<code>docs/MIMIC_FLOWS_COMPLETE_GUIDE.md</code>	Visual-first still described as TP-grounded; missing <code>execution_mode</code> , Why Mimic, new visual, BVS
<code>docs/CREATIVE_INTELLIGENCE.md</code>	No SIL, <code>mimic_evaluation</code> , new-visual lane
<code>docs/MIMIC_IMAGE_FLOWS.md</code>	Pre-BVS, pre-new-visual
<code>visual-first-carousel-flow-types.ts</code> header comment	Still says TP-grounded references
<code>docs/CAF_CORE_COMPLETE_GUIDE.md</code>	Merged summary stale on BVS, Why Mimic, new visual (June 2026 stamp)
<code>docs/DATABASE_SCHEMA.md</code>	Missing <code>brand_bibles</code> , <code>brand_profiles</code> , <code>job_outcomes</code> , slide intelligence columns
<code>docs/export/README.md</code>	Last generated 2026-06-16; no current-state pack

docs/JOB_LIFECYCLE.md	May omit GENERATED, QC_FAILED split-brain
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Features in code, not documented

- Brand Visual System (brand_bibles , bvs_v1 , bvs_render_plan , invented plates)
- FLOW_WHY_MIMIC_CAROUSEL + slide intelligence layer
- FLOW_VISUAL_FIRST_CAROUSEL as new visual (not TP replication)
- execution_mode on mimic_v1 (classic | why_mimic | new_visual)
- job_outcomes table (075)
- CarouselBrandStylingPanel + live preview overlays in Review
- Per-project mimic render settings (migrations 066–069)
- MIMIC_CAROUSEL_TEXT_VIA_FLUX ignored at render (hardcoded false)

Documented but not fully implemented

- docs/MIMIC_TEXT_PLACEMENT_AUTOMATION.md — composite saliency analyzer
- Project risk_rules enforcement (documented as config-only — correct)
- Stage-3 structured idea picker (roadmap TBD)
- Global learning rules in generation compiler

Suggested update priority

1. docs/MIMIC_FLOWS_COMPLETE_GUIDE.md — add execution modes, new visual, Why Mimic, BVS section
2. docs/DATABASE_SCHEMA.md — brand_bibles, job_outcomes, SIL columns
3. docs/CREATIVE_INTELLIGENCE.md — downstream SIL + mimic_evaluation
4. docs/CAF_CORE_COMPLETE_GUIDE.md — merge BVS + new flows or point to this pack
5. docs/export/README.md — add current-state PDF volumes
6. Cursor rule visual-first-carousel-flow.mdc — align header with new_visual semantics
7. docs/JOB_LIFECYCLE.md — add GENERATED , clarify QC_FAILED scope

Appendix A — ChatGPT upload bundle (2026-07)

For external LLM projects, upload in order:

Priority	File
1	docs/CAF_CURRENT_STATE_CONTEXT_PACK.md (this file) OR volumes 1–4
2	AGENTS.md
3	docs/EXTERNAL_CONTEXT_PACK.md
4	apps/review/CHATGPT_CAROUSEL_REVIEW_CONSOLE_EDIT_GUIDE.md (operator mimic workflow)
5	Topic add-ons: docs/MIMIC_FLOWS_COMPLETE_GUIDE.md (until updated), docs/CREATIVE_INTELLIGENCE.md

System prompt: Use the template in docs/EXTERNAL_CONTEXT_PACK.md § "System prompt template".

Regenerate PDFs: npm run export:doc-pdfs after doc changes.